Cell Tuition
1-1 PRIVATE TUTORS

YEAR 9 · ALGEBRA

Collecting Like Terms

Name _____ **Date** _____ No calculator Version 1

KEY WORDS term · like terms · coefficient · variable · simplify · expression

A One letter ●○○ — ANSWER ON THE SHEET

Remember: Like terms have exactly the same letters. Add or subtract the numbers in front: $3x + 5x = 8x$.

1. Simplify.

a) $3x + 5x =$ _____ **b)** $7a - 2a =$ _____ **c)** $4y + y =$ _____ **d)** $9m - 9m =$ _____

e) $2x + 3x + 4x =$ _____ **f)** $8p - 3p + p =$ _____ **g)** $5k - 8k =$ _____ **h)** $-2n + 7n =$ _____

i) $6b + 6b - 5b =$ _____ **j)** $-3t - 4t =$ _____ **k)** $10w - w - 6w =$ _____ **l)** $c + c + c =$ _____

m) $2d + 9d =$ _____ **n)** $12z - 7z =$ _____ **o)** $3h + 3h + 3h =$ _____ **p)** $-5r + 5r =$ _____

q) $15x - 6x - 4x =$ _____ **r)** $q + 8q =$ _____ **s)** $-v - v =$ _____ **t)** $20e - 11e =$ _____

B Two letters ●●○ — ANSWER ON THE SHEET

Watch out: Terms with different letters cannot be added: $3x + 2y$ stays as it is.

2. Simplify by collecting like terms.

a) $3x + 2y + 4x =$ _____ **b)** $5a + 3b - 2a =$ _____ **c)** $6m + 4n + 2n =$ _____

d) $7p - 3q - 4p =$ _____ **e)** $2x + 5y + 3x + y =$ _____ **f)** $8a + 2b - 3a - 5b =$ _____

g) $4c - 6d + 2c + 9d =$ _____ **h)** $x + y + x + y =$ _____ **i)** $9h - 2k - 9h + 5k =$ _____

j) $-3a + 4b + 7a - 6b =$ _____ **k)** $5x - 2y - 8x + 3y =$ _____ **l)** $6u + 6v - 6u - v =$ _____

m) $10x - 3y - 4x - 2y =$ _____ **n)** $2s + 7t + 5s - 7t =$ _____ **o)** $a - b + a - b =$ _____

p) $12m + 5n - 15m + n =$ _____ **q)** $3x + 3y + 3x - 3y =$ _____ **r)** $-4g - 2h + 9g + 6h =$ _____

C Terms with powers and numbers ●●○ — WORK IN YOUR BOOK

Watch out: x^2 and x are not like terms. A number on its own can only be added to another number.

3. Simplify.

a) $3x^2 + 2x^2$ **b)** $5x + 4 + 2x + 3$ **c)** $4x^2 + 3x + 2x^2$

d) $7a + 9 - 2a - 5$ **e)** $6y^2 - 2y - 4y^2 + 5y$ **f)** $2x^2 + 3x + 1 + 4x^2 - x$

g) $10 + 3m - 4 + 2m$ **h)** $a^2 + a + a^2 + a$ **i)** $5p^2 - 5p - 2p^2 - 3p$

j) $8x^2 - 3x^2 + x^2$ **k)** $2t^2 + 6t + 5 - 2t^2$ **l)** $9 + 4k - 9 - 4k$

m) $3n^2 + 7 + 2n - n^2 - 10$ **n)** $x^3 + 2x^2 + 3x^3$ **o)** $5w - 2w^2 + 3w + 6w^2$

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D Mixed practice ●●● — WORK IN YOUR BOOK

4. Simplify.

a) $3ab + 2ab$

b) $4xy + 3x + 2xy$

c) $5a + 2b + 3c + a - 2c$

d) $7x - 3y + 4 - 2x + 5y - 1$

e) $2mn + 5m - mn - 5m$

f) $6a^2 + 2ab - 3a^2 + 4ab$

g) $0.5x + 1.5x + 2y$

h) $9p - 4q - 9p + 4q$

i) $3x^2 - x + 8 - 3x^2 + 6x - 8$

j) $2xy + 3yz + 4xy - yz$

k) $5a - 2ab + 3b + 2ab$

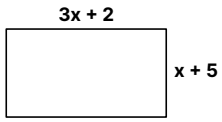
l) $x^2y + 4xy + 2x^2y$

E Perimeter ●●● — WORK IN YOUR BOOK

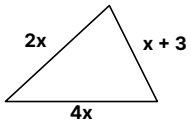
> **Remember:** The perimeter is the distance around a shape. Add all the sides, then collect like terms.

5. Write an expression for the perimeter of each shape. Simplify your answer.

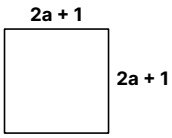
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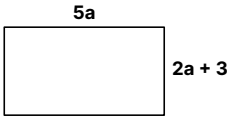
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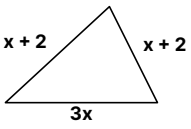
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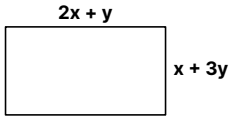
d)



e)



f)



★ Extension — FOR EARLY FINISHERS

6. Simplify $3x + 2y - (1x - 4y)$. Take care with the signs inside the bracket.

7. A rectangle has a perimeter of $10x + 6$. Its length is $3x + 1$. Find an expression for its width.

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Collecting Like Terms

Name _____ **Date** _____ No calculator **Version 2**

KEY WORDS term · like terms · coefficient · variable · simplify · expression

A One letter ●○○ — ANSWER ON THE SHEET

Remember: Like terms have exactly the same letters. Add or subtract the numbers in front: $3x + 5x = 8x$.

1. Simplify.

a) $4x + 2x =$ _____ **b)** $9a - 4a =$ _____ **c)** $6y + y =$ _____ **d)** $5m - 5m =$ _____

e) $3x + 2x + 5x =$ _____ **f)** $7p - 2p + p =$ _____ **g)** $4k - 9k =$ _____ **h)** $-3n + 8n =$ _____

i) $5b + 5b - 4b =$ _____ **j)** $-2t - 6t =$ _____ **k)** $12w - w - 5w =$ _____ **l)** $d + d + d =$ _____

m) $3c + 8c =$ _____ **n)** $14z - 6z =$ _____ **o)** $4h + 4h + 4h =$ _____ **p)** $-7r + 7r =$ _____

q) $16x - 5x - 7x =$ _____ **r)** $q + 9q =$ _____ **s)** $-u - u =$ _____ **t)** $25e - 13e =$ _____

B Two letters ●●○ — ANSWER ON THE SHEET

Watch out: Terms with different letters cannot be added: $3x + 2y$ stays as it is.

2. Simplify by collecting like terms.

a) $4x + 3y + 2x =$ _____ **b)** $6a + 2b - 3a =$ _____ **c)** $5m + 3n + 4n =$ _____

d) $8p - 2q - 5p =$ _____ **e)** $3x + 4y + 2x + y =$ _____ **f)** $9a + 3b - 4a - 7b =$ _____

g) $5c - 8d + 3c + 10d =$ _____ **h)** $a + b + a + b =$ _____ **i)** $7h - 3k - 7h + 6k =$ _____

j) $-2a + 5b + 9a - 8b =$ _____ **k)** $4x - 3y - 9x + 5y =$ _____ **l)** $8u + 8v - 8u - v =$ _____

m) $11x - 2y - 5x - 3y =$ _____ **n)** $3s + 8t + 4s - 8t =$ _____ **o)** $c - d + c - d =$ _____

p) $10m + 4n - 14m + n =$ _____ **q)** $5x + 5y + 5x - 5y =$ _____ **r)** $-3g - 4h + 8g + 9h =$ _____

C Terms with powers and numbers ●●○ — WORK IN YOUR BOOK

Watch out: x^2 and x are not like terms. A number on its own can only be added to another number.

3. Simplify.

a) $4x^2 + 3x^2$ **b)** $6x + 2 + 3x + 5$ **c)** $5x^2 + 2x + 3x^2$

d) $8a + 7 - 3a - 4$ **e)** $7y^2 - 3y - 5y^2 + 6y$ **f)** $3x^2 + 4x + 2 + 2x^2 - x$

g) $12 + 4m - 5 + 3m$ **h)** $b^2 + b + b^2 + b$ **i)** $6p^2 - 4p - 3p^2 - 2p$

j) $9x^2 - 4x^2 + x^2$ **k)** $3t^2 + 5t + 4 - 3t^2$ **l)** $7 + 6k - 7 - 6k$

m) $4n^2 + 5 + 3n - n^2 - 9$ **n)** $x^3 + 4x^2 + 2x^3$ **o)** $6w - 3w^2 + 2w + 5w^2$

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D Mixed practice ●●● WORK IN YOUR BOOK

4. Simplify.

a) $4ab + 3ab$

b) $5xy + 2x + 3xy$

c) $6a + 3b + 2c + a - c$

d) $8x - 2y + 5 - 3x + 6y - 2$

e) $3mn + 4m - 2mn - 4m$

f) $7a^2 + 3ab - 2a^2 + 5ab$

g) $1.5x + 2.5x + 3y$

h) $7p - 5q - 7p + 5q$

i) $4x^2 - x + 6 - 4x^2 + 7x - 6$

j) $3xy + 2yz + 5xy - yz$

k) $6a - 3ab + 2b + 3ab$

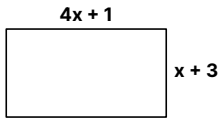
l) $x^2y + 5xy + 3x^2y$

E Perimeter ●●● WORK IN YOUR BOOK

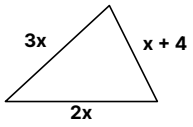
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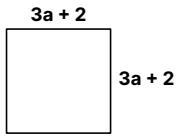
a)



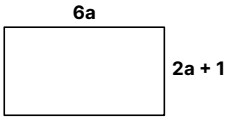
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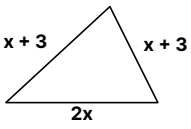
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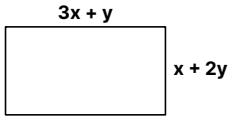
d)



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★ Extension FOR EARLY FINISHERS

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